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Optimization of tuned mass dampers for multiple mode vortex-induced vibration mitigation in flexible structures: An application to multi-span continuous bridge

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ABSTRACT

Tuned mass dampers (TMDs) have proven effective in suppressing structural vibration. In existing research on vortex-induced vibration (VIV) control of flexible structures, only one mode is commonly considered at the time, and only basic aerodynamic force models, which can struggle to model amplitude-dependent damping, are considered. However, flexible structures, such as multi-span bridges, exhibit multiple modes with closely spaced frequencies. Neglecting secondary modes may introduce calculation inaccuracies, as these modes can impose additional stiffness, thereby influencing the performance of TMD. This paper presents a framework to optimize the performance of multiple TMDs in the context of multi-mode VIV control for a flexible structure. A multi-span bridge is used as a case study, and the secondary modes and nonlinear aerodynamic effects are considered. The governing equations for the force-structure-MTMDs system are derived in modal coordinates using a polynomial model to model the aerodynamic force. A novel method to calculate the contribution of each mode to determine the minimum number of secondary modes to be considered is proposed. The optimal parameters of MTMDs are obtained using a global optimization method and compared with mode-by-mode design results. Robustness analysis is conducted to verify the effectiveness of the optimization framework. The results show that the polynomial model accurately predicts the amplitude after TMD installation. Secondary modes significantly affect the optimal parameters of TMDs, and the optimized results exhibit better control performance than the mode-by-mode design results.

1. Introduction

Recent advancements in civil engineering have enabled the development of structures situated in deeper oceanic waters, at higher altitudes, and featuring greater spans to address the diverse needs of the population. The advancement of innovative materials, large-scale structures, and construction methodologies has resulted in slender structures characterized by decreased structural rigidity and damping ratios [1]. Notably, the slender and flexible characteristics of long-span bridges [1–10], high-rise buildings [11,12], chimneys [13,14], wind turbines [15] and extensive marine riser pipes [16,17] make them susceptible to many

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Table 1
Flexible structures with VIV.

Flexible structure	Type	Length (m)	Frequency (Hz)					Ref.
			Mode 1	Mode 2	Mode 3	Mode 4	Mode 5	
APPR of Great Belt East Bridge	CONT	193	0.49[a]	0.54[a]	0.61[a]	0.70[a]	0.80[a]	[3]
Rio-Niterói bridge	CONT	300	0.32	0.55	/	/	/	[4]
Trans-Tokyo Bay Crossing Bridge	CONT	240	0.33	0.47	0.61	0.65	0.72	[6]
APPR of Lingding Yang Birdge	CONT	110	0.83	0.90	1.06	1.28	1.51	This paper
Alconétar Bridge	ARCH	220	0.7	/	/	/	/	[7]
Great Belt East Bridge	SUSP	1624	0.10	0.13	0.17	0.21	0.24	[5]
Yi Sun-sin Bridge	SUSP	1545	0.11	0.15	0.22	0.24	0.31	[59]
Xihoumen Bridge	SUSP	1650	0.10	0.13	0.18	0.23	0.28	[60]
Stay-cable	/	197.9	0.57	/	/	/	/	[28]
Hanger	/	127	0.90[b]	/	/	/	/	[29]

Note: APPR: Approach span, CONT: Continuous bridge, ARCH: Arch bridge, SUSP: Suspension bridge.

[a] : Mean.

[b] : Approx value.

kinds of vibrations. The categorization of vibrations reveals that the most prevalent forms encompass fluid-induced vibrations [6,18], seismic vibrations [19–22], and those induced by pedestrians and vehicles [23–25]. A particularly intricate aspect is the regulation of fluid-induced vibrations, specifically vortex-induced vibrations (VIV), attributed to the nonlinear interaction between the structure and fluid [26]. This nonlinearity renders conventional control approaches, which rely on frequency-domain analysis and linear control theory, unsuitable. Nevertheless, it is imperative to note that VIV is a recurring phenomenon in the aforementioned structures. Furthermore, specific components of these structures, such as cables [27,28] and hangers [29] in suspension bridges, along with wind turbine blades [30,31], are also prone to VIV. Although VIV may not lead to an immediate structural collapse, it can result in discomfort for the occupants and contribute to the long-term fatigue damage of the structural materials [32].

Previous studies have described two main approaches to suppress VIV: fluid dynamic [33,34] and mechanical countermeasures [35,36]. Of the two, fluid dynamic countermeasures are often considered cost-effective, achieved by tailoring the structure's shape or installing auxiliary structure facilities. Consequently, these countermeasures have become widely utilized and are typically the first choice for VIV control [1,37–39]. Nevertheless, determining the optimal design for the configuration of these structures remains a complex task, and the underlying mechanisms governing the interactions between fluid and the structures have not yet been thoroughly examined. As a result, mechanical countermeasures represent a promising alternative because they effectively control VIV regardless of the structures' shape [40]. The most popular mechanical countermeasure is a tuned mass damper (TMD), which consists of a mass block, a spring and a damping element [3,41]. How to use TMD to control the vibration of a single degree of freedom structure is now well known [42]. The design principles for a TMD attached to a single-DoF structure under harmonic load have been thoroughly explored, both in absolute and modal coordinates [43–47]. The assumption of a single-DoF vibration is commonly endorsed for the control of VIV in high-rise buildings, towers, and chimneys [41]. This widespread acceptance is attributed to the fact that the frequencies associated with higher modes are much larger than the fundamental frequency of the structure [48,49]. Consequently, these higher modes are less prone to the excitation by wind forces and exhibit reduced coupling effects with the fundamental mode [50]. Some studies use multiple TMDs (MTMDs) to replace the single TMD for a target control mode to improve the control robustness [51,52]

However, for many structures, the vibration is inherently a multi-mode phenomenon [17,53–55]. Table 1 presents the natural frequencies of several flexible structures reported to have multi-mode VIV events. The complexity of multi-mode VIV is often compounded by the structures' tendency to exhibit closely spaced frequencies. This characteristic implies an increase in the number of lock-in regions within the reference wind speed. Typically, flexible structures exhibit closely spaced frequencies, where adjacent frequencies generally fall within a range of approximately 10% of a shared average, although this percentage can vary based on specific criteria [56,57]. Fig. 1 shows the relationship between the frequency of the first mode and subsequent modes by normalizing the frequencies with the frequency of the first mode for several birdges. The figure also shows results for a flexible inclined stay-cable [58] with closely spaced frequencies. Square markers represent suspension bridges, circles indicate continuous bridges and diamond markers symbolize the flexible stay-cable. Despite possessing an elevated fundamental frequency, continuous bridges demonstrate a dense frequency distribution, wherein neighboring frequencies remain within a 10% deviation from their average frequency.

In addressing the vibration control challenge presented by structures with closely spaced frequencies, Krenk and Høgsberg [61] emphasized the importance of considering secondary modes, as they offer additional flexibility. However, the vibration is inherently a multimode phenomenon for actual structures once dampers are installed, making it challenging to derive a closed-form solution through theoretical analysis without making some assumptions [62–64]. By contrast, a numerical method may be more practical for addressing the complex interplay of coupled modes and TMDs [65–67]. Using optimization algorithms to find the optimal parameters of TMDs is a promising approach [68–70]. The amalgamation of the system matrix with MTMDs is conventionally accomplished through two primary methods. The first approach entails the direct integration of the structural matrix, a technique characterized by superior accuracy albeit at the expense of heightened computational demands. As a result, this technique is only suitable for certain structures that can be accurately simplified [21,22], rendering it less than ideal for tackling optimization issues in actual engineering structures. Alternatively, modal reduction can be employed, which involves retaining only the first modes

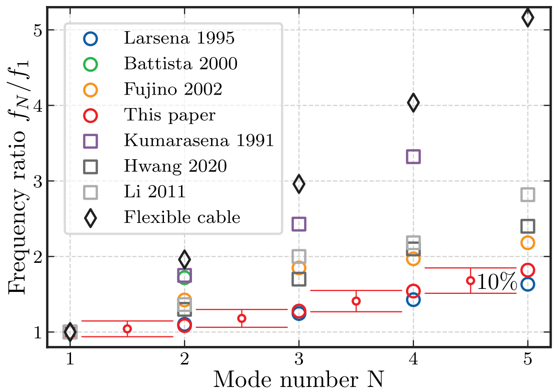


Fig. 1. Variation of the frequency ratios of long-span bridges.

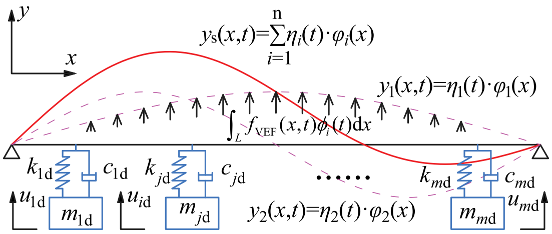


Fig. 2. Schematic diagram of the force-structure-MTMDs system.

of the direct integration of the structural matrix [68,71–75]. Most studies typically match the number of TMDs with the number of resonant modes and the reduced structural modes. These TMDs are strategically placed at locations where the target modal displacement is at its maximum [75]. This implies that, except for the resonant modes, all other secondary modes of the structure are disregarded. Particularly for structures characterized by widely spaced frequencies, this principle holds. However, for structures with closely spaced frequencies, the interaction between modes caused by TMDs is non-negligible, particularly concerning those secondary modes devoid of resonant vibrations. Thus, the number of the reduced structural modes taken into calculation should be carefully considered. While there are some studies on optimizing MTMDs for multiple-mode VIV of bridge structures [76,77], the interactions between the modes have often been disregarded. While an array of innovative dampers has been developed over the past two decades to mitigate VIV [76,78–90], the impact of secondary modes on VIV control, particularly for structures exhibiting closely spaced frequencies, remains insufficiently investigated. Regardless of the type of damper used, as long as VIV occurs in a structure with closely spaced frequencies, vibration energy will transfer from the dominant mode to other modes via the coupled motion of the structure and the damper. It is essential to determine whether ignoring the interaction between modes would significantly affect the accuracy of the calculated response.

Previous research on VIV control has primarily included one mode in the modeling and applied a simplified aerodynamic force model, potentially leading to imprecise estimates of the aerodynamic or hydrodynamic damping ratio. To the authors' knowledge, the influence of secondary modes on the calculation accuracy of VIV control using TMDs for structures with closely spaced frequencies has not been comprehensively investigated, leaving a gap in the availability of quantitative analysis. The primary innovation of this investigation resides in the development of a framework to design MTMDs for structures exhibiting closely spaced frequencies, incorporating secondary modes and the nonlinearity of the force model. The proposed computational approach is capable of delivering enhanced accuracy in determining the VIV amplitude and the vibration control efficacy of MTMDs. A 660 m six-span continuous bridge is selected as the case study to demonstrate the effectiveness of the proposed framework. The proposed framework is not only restricted to bridges, it can also be applied to other structures with multiple modes. The principal objective of this research is to ascertain the minimal number of modes necessary to optimize the MTMD parameters for VIV control. The paper is organized as follows. In Section 2, the governing equations of the force-structure-TMDs dynamic system, the numerical calculation method, and the optimization framework are introduced. In Section 3, the VIV response calculation employing MTMDs and a polynomial force model is validated through comparison with experimental results. Subsequently, the impact of secondary modes on the calculation accuracy for VIV responses in bridges exhibiting closely spaced frequencies is thoroughly examined. The optimization results for the TMD parameters are presented in Section 4, which show a better performance than the mode-by-mode design. Finally, the conclusions are presented in Section 5.

2. Governing equations and optimization

2.1. Modeling of force-structure-MTMDs system

A simply supported beam with multiple TMDs is considered. Fig. 2 depicts the layout of the system, where x and y denote the spanwise and vertical coordinates, respectively. Only vertical aerodynamic forces and vibrations are considered. The mass element of each TMD is connected to the structure through a linear spring and dashpot damper. The displacement of the structure can be described as a combination of multiple modes using the superposition method, which is illustrated in the figure as the red solid and pink dashed lines.

The damping force and stiffness force of the jth TMD are expressed as

$$F_{j\mathrm{c}}(t) = c_{j\mathrm{d}}\left[\frac{\partial y_{\mathrm{s}}\left(x_j,t\right)}{\partial t} - \frac{\partial y_{j\mathrm{d}}(t)}{\partial t}\right] \tag{1}$$

$$F_{j\mathrm{k}}(t) = k_{j\mathrm{d}}\left[y_{\mathrm{s}}\left(x_j,t\right) - y_{j\mathrm{d}}(t)\right] \tag{2}$$

where $c_{j\mathrm{d}}$ and $k_{j\mathrm{d}}$ denote the damping and stiffness coefficients of the jth TMD, and $y_{j\mathrm{d}}(t)$ and $y_{\mathrm{s}}\left(x_j,t\right)$ represent the displacement of the jth TMD and the structure at the TMD location.

The equation of motion for the force-structure-MTMDs system is derived using the principle of virtual work. The motion of the structure is shown in Eq. (3), and the motion of each TMD is shown in Eq. (4).

$$\begin{aligned}&\int_L \left[m_{\mathrm{s}}(x)\frac{\partial^2 y_{\mathrm{s}}(x,t)}{\partial t^2} + c_{\mathrm{s}}(x)\frac{\partial y_{\mathrm{s}}(x,t)}{\partial t} + \mathrm{EI}(x)\frac{\partial^4 y_{\mathrm{s}}(x,t)}{\partial x^4}\right]\cdot \delta y_{\mathrm{s}}(x,t)dx\\ &+\sum_j\left[F_{j\mathrm{c}}(t) + F_{j\mathrm{k}}(t)\right]\cdot \delta y_{\mathrm{s}}\left(x_j,t\right) = \int_L f_{\mathrm{VEF}}(x,t)\cdot \delta y_{\mathrm{s}}(x,t)dx\end{aligned} \tag{3}$$

$$\left[m_{j\mathrm{d}}\ddot{u}_{jd}(t) - F_{j\mathrm{c}}(t) - F_{j\mathrm{k}}(t)\right]\cdot \delta u_{jd}(x,t) = 0 \tag{4}$$

where $m_{\mathrm{s}}(x)$, $c_{\mathrm{s}}(x)$, and $\mathrm{EI}(x)$ represent the distributed mass, distributed damping coefficient, and bending stiffness of the structure while $m_{j\mathrm{d}}$, $F_{j\mathrm{c}}(t)$, $F_{j\mathrm{k}}(t)$ are the mass element, damping force and stiffness force of the TMD. The symbol $y_{\mathrm{s}}(x,t)$ represents the displacement of the structure at the location of x, $\ddot{u}_{jd}(t)$ is the acceleration of TMD, and $\delta y_{\mathrm{s}}(x,t)$ and $\delta u_{j\mathrm{d}}(x,t)$ are the virtual displacement of structure and TMD. $f_{\mathrm{VEF}}(x,t)$ represents the aerodynamic force acting on the structure.

The structural and virtual displacements can be decomposed into the summation of modal displacements using the superposition principle given by Eq. (5).

$$\begin{cases} y_{\mathrm{s}}(x,t) = \sum_n \phi_i(x)\eta_i(t)\\ \delta y_{\mathrm{s}}(x,t) = \sum_n \delta\phi_i(x)\eta_i(t)\end{cases} \tag{5}$$

where $\phi_i(x)$ represents the modal shape of the ith mode, and $\eta_i(t)$ represents the time history of modal displacement of the ith mode.

It should be highlighted that for a simply supported beam, the efficiency of the modal superposition method may not exhibit a marked improvement. However, the scenario changes when dealing with the structure of an actual project, which is invariably more complex, encompassing a greater number of structural components and varied sectional configurations along the span. This complexity introduces additional degrees of freedom into the structure, enhancing the efficiency of the modal superposition method when the first modes of the structure are retained. It is also important to clarify that the modal information utilized in Eq. (5) is derived from the structure in the absence of TMDs. This approach is necessitated by the fact that incorporating TMDs alters the modes of the structure, complicating the identification of resonant VIV modes. Consequently, an underlying assumption of our analysis is that the TMDs exert a negligible influence on the structural modes. To address the potential calculation inaccuracies resulting from neglect, our approach involves incorporating a multitude of secondary modes into the calculation, as delineated in Section 3.5.

By substituting Eq. (5) into Eq. (3), the governing equations can be derived as follows:

$$\begin{aligned}&\int_L\left[\begin{matrix} m_{\mathrm{s}}(x)\sum_n \phi_i(x)\ddot{\eta}_i(t) + c_{\mathrm{s}}(x)\sum_n \phi_i(x)\dot{\eta}_i(t)\\ +\mathrm{EI}(x)\sum_n \phi''''_i(x)\eta_i(t)\end{matrix}\right]dx\\ &+\sum_j\left[F_{j\mathrm{c}}(t) + F_{j\mathrm{k}}(t)\right] = \int_L f_{\mathrm{VEF}}dx\end{aligned} \tag{6}$$

$$m_{j\mathrm{d}}\ddot{u}_{jd}(t) - F_{j\mathrm{c}}(t) - F_{j\mathrm{k}}(t) = 0 \tag{7}$$

The equation of the ith mode can be obtained by multiplying $\phi_i(x)$ in Eq. (6) and integrating it over the length of the structure.

$$\begin{aligned}&m_i\ddot{\eta}_i(t) + c_i\dot{\eta}_i(t) + k_i\eta_i(t)\\ &+\sum_j\left[F_{j\mathrm{c}}(t) + F_{j\mathrm{k}}(t)\right]\cdot\phi_i\left(x = x_j\right) = \int_L f_{\mathrm{VEF}}(x,t)\phi_i(x)dx\end{aligned} \tag{8}$$

Matrix form can be used to express the dynamic equations of the force-structure-MTMDs system, considering the first n modes of the structure and m TMDs, as follows:

$$\mathbf{M}\cdot\ddot{\boldsymbol{\eta}}+\mathbf{C}\cdot\dot{\boldsymbol{\eta}}+\mathbf{K}\cdot\boldsymbol{\eta}=\mathbf{F}_{\mathrm{VEF}}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}}) \tag{9}$$

where $\mathbf{M}$, $\mathbf{C}$, and $\mathbf{K}$ are the mass, damping, and stiffness matrices of the system, as shown in Eqs. (10)–(12), respectively.

$$\mathbf{M}=\mathrm{diag}\left(m_1,\ldots,m_n,m_{1d},\ldots,m_{md}\right) \tag{10}$$

$$\mathbf{C}=\begin{bmatrix} c_1+\sum_{\mathrm{j}=1}^{m} c_{jd}\phi^2{}_{x=xj,1} & \cdots & \sum_{\mathrm{j}=1}^{m}\left[c_{jd}\phi_{x=xj,n}\right]\phi_{x=xj,1} & -c_{1d}\phi_{x=x1,1} & \cdots & -c_{md}\phi_{x=xm,1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \sum_{\mathrm{j}=1}^{m}\left[c_{jd}\phi_{x=xj,1}\right]\phi_{x=xj,n} & \cdots & c_n+\sum_{\mathrm{j}=1}^{m} c_{jd}\phi^2{}_{x=xj,n} & -c_{1d}\phi_{x=x1,n} & \cdots & -c_{md}\phi_{x=xm,n} \\ -c_{1d}\phi_{x=x1,1} & \cdots & -c_{1d}\phi_{x=x1,n} & c_{1d} & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ -c_{md}\phi_{x=xm,1} & \cdots & -c_{md}\phi_{x=xm,n} & 0 & \cdots & c_{md} \end{bmatrix} \tag{11}$$

$$\mathbf{K}=\begin{bmatrix} k_1+\sum_{\mathrm{j}=1}^{m} k_{jd}\phi^2{}_{x=xj,1} & \cdots & \sum_{\mathrm{j}=1}^{m}\left[k_{jd}\phi_{x=xj,n}\right]\phi_{x=xj,1} & -k_{1d}\phi_{x=x1,1} & \cdots & -k_{md}\phi_{x=xm,1} \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ \sum_{\mathrm{j}=1}^{m}\left[k_{jd}\phi_{x=xj,1}\right]\phi_{x=xj,n} & \cdots & k_n+\sum_{\mathrm{j}=1}^{m} k_{jd}\phi^2{}_{x=xj,n} & -k_{1d}\phi_{x=x1,n} & \cdots & -k_{md}\phi_{x=xm,n} \\ -k_{1d}\phi_{x=x1,1} & \cdots & -k_{1d}\phi_{x=x1,n} & k_{1d} & \cdots & 0 \\ \vdots & \ddots & \vdots & \vdots & \ddots & \vdots \\ -k_{md}\phi_{x=xm,1} & \cdots & -k_{md}\phi_{x=xm,n} & 0 & \cdots & k_{md} \end{bmatrix} \tag{12}$$

$$\boldsymbol{\eta}=\begin{bmatrix} \eta_1 & \cdots & \eta_n & u_{1d} & \cdots & u_{md} \end{bmatrix}^T \tag{13}$$

where the vector $\boldsymbol{\eta}$ contains both the modal displacement of the structure and the actual displacement of the TMDs.

$$\mathbf{F}_{\mathrm{VEF}}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}})=\mathbf{S}\cdot\int_L f_{\mathrm{VEF}}(x,t)\phi_i(x)dx\cdot \tag{14}$$

where $\mathbf{S}\in\mathbb{R}^{(n+m)\times 1}$ is the binary matrix that selects the degree of freedom where vortex-excited force is added. $\mathbf{F}_{\mathrm{VEF}}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}})$ is the modal vortex excited forces.

The equations reveal that the vibration is excited by the vortex-excited force for a certain mode, after which the TMDs begin to vibrate to reduce the vibration. However, the reaction force of the TMD to the structure affects all the modes that are considered for the structure. As a result, the vibration energy is transferred from the resonant mode to the other secondary modes due to the interaction between the structure and the TMDs. Solving these equations yields results that account for the influence of the secondary modes, which can be crucial to obtain accurate results. It is important to acknowledge that, while the governing equations are formulated based on the simply supported beam model for the sake of straightforward demonstration, the proposed methodology retains its applicability to more complex structural types, such as continuous and suspension bridges. Moreover, although the calculation model is founded on the vertical modes, the proposed methodology is equally applicable to the torsional modes, provided that the torsional aerodynamic forces are accurately identified and the corresponding torsional modes are integrated into the governing equations. However, it is crucial to underline that there has been no recorded observation of torsional VIV in the field, hence its exclusion from consideration in this study [32]. Furthermore, owing to the constraints of the existing aerodynamic force model, the applicability of the proposed methodology is restricted to the analysis of VIV for a singular mode of the structure at any given instance. This constraint arises from the fact that the polynomial model is formulated to handle vibrations at a singular frequency. Strictly speaking, the aerodynamic force model is not designed to calculate VIV involving multiple frequencies simultaneously. However, it is noteworthy that VIV in bridges is typically governed by a single dominant mode at specific wind speeds. This particular behavior provides a rationale for the adoption of the polynomial model in this study, especially when no more suitable alternatives are available as demonstrated in Section 2.2.

2.2. VIV-induced nonlinear aerodynamic force model

VIV exhibits significant nonlinear characteristics, such as self-excitation and self-limiting response [26,91]. Since the TMDs can alter the damping ratio of the system and thus affect the amplitude of the VIV, it is crucial that the aerodynamic force model accurately models the vortex-excited force at different Scruton numbers. This study employs a polynomial model as recommended by Zhang et al. [92]. It can accurately model the aerodynamic damping ratio at different amplitudes and Scruton numbers, as shown in Eq. (15).

$$f_{\mathrm{VEF}}(y,\dot{y})=\rho U^2 D\left[\sum_{k=1}^{m} a_{2k-1}\left(\frac{y}{D}\right)^{2(k-1)}\frac{\dot{y}}{U}+\sum_{l=1}^{n} a_{2l}\left|\frac{y}{D}\right|^{2l-1}\frac{\dot{y}}{U}\right] \tag{15}$$

where ρ is the air density, D represents the reference length, U is the wind speed, y is the vertical displacement, and a_{2k-1} (where k ranges from 1 to m) and a_{2l} (where l ranges from 1 to n), which are aerodynamic parameters dependent on the reduced frequency $K = \omega_0 D/U$ where ω_0 is the natural frequency of the structure.

The first five terms of the polynomial model are used in this study, using $m = 3$ and $n = 2$. By substituting Eq. (15) into Eq. (14), the vortex-excited force vector can be expressed in modal coordinates, as shown in Eq. (16).

$$\mathbf{F}_{\mathrm{VEF}}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}}) = \mathbf{S}\cdot\rho U^2 D\begin{bmatrix} a_1\int_0^l \phi_i^2(x)dx + a_2\int_0^l \phi_i^3(x)dx\left|\frac{\eta_i}{D}\right| + a_3\int_0^l \phi_i^4(x)dx\left(\frac{\eta_i}{D}\right)^2 \\ +a_4\int_0^l \phi_i^5(x)dx\left|\frac{\eta_i}{D}\right|^3 + a_5\int_0^l \phi_i^6(x)dx\left(\frac{\eta_i}{D}\right)^4 \end{bmatrix}\frac{\dot{\eta}_i}{U} \tag{16}$$

The equation reveals that the first term, $\rho U^2 D a_1\int_0^l \phi_i^2(x)dx\frac{\dot{\eta}_i}{U}$, is a linear damping term, while the other terms contribute are nonlinear damping and depends on the modal displacement η_i. Consequently, when η_i is zero, only the linear term persists. The heavy computational burden of time histories can be reduced to an eigenvalue analysis, which allows for evaluating the system's stability at zero amplitude of VIV.

2.3. Numerical solution for dynamic system

2.3.1. Linear stability analysis

As previously mentioned, solving the nonlinear differential equation in Eq. (9) analytically is challenging. Thus, this study uses two numerical solutions to solve the dynamic system. Complex eigenvalue analysis provides the damping ratio of the system at a specific amplitude and can be used to determine whether VIV will occur. A negative damping ratio indicates instability in the structure-TMD system at the onset of VIV. The eigenvalue analysis is based on the linearized equation of motion at a specific amplitude. The linearized aerodynamic force model vector at zero amplitude is shown in Eq. (17).

$$\begin{aligned}\mathbf{F}_{\mathrm{VEF}}{}^*(\dot{\boldsymbol{\eta}}) &= \mathbf{S}\cdot F_{\mathrm{VEF}}^*(\dot{\eta}) \\ &= \mathbf{S}\cdot\rho U^2 D a_1\int_0^l \phi_i^2(x)dx\frac{\dot{\eta}_i}{U}\end{aligned} \tag{17}$$

The linearized vortex-excited force vector is only dependent on the modal velocity, allowing it to be combined with the structural damping matrix to form a new damping matrix $\mathbf{C}^*$.

$$\mathbf{C}^* = \mathbf{C} - \mathbf{F}_{\mathrm{VEF}}{}^*(\dot{\boldsymbol{\eta}})/\dot{\boldsymbol{\eta}} \tag{18}$$

The complex eigenvalues of the force-structure-MTMDs system can be obtained by solving the eigenvalue problem in Eq. (19). This equation represents a standard eigenproblem of order $2n$, with eigenvalues appearing in complex conjugate pairs for underdamped systems. The natural frequencies and damping ratios can be obtained by solving Eq. (19). The mode with the lowest damping ratio is the most unstable, and the corresponding eigenvector is the mode shape. The stable amplitude can also be calculated by linearizing the nonlinear aerodynamic damping ratio at different amplitudes and using an iterative method to control the lowest damping ratio to zero. In this study, the eigenvalue analysis is only used to determine the system's stability. The displacement is calculated using the nonlinear Newmark-beta method, which is more intuitive and will be described in the following section.

$$\begin{bmatrix} -\mathbf{M}^{-1}\mathbf{C}^* & \mathbf{M}^{-1}\mathbf{K} \\ \mathbf{I} & 0 \end{bmatrix}\mathbf{V} = \mathbf{s}\mathbf{V} \tag{19}$$

$$\boldsymbol{\omega}_{\mathrm{s}} = -\,\mathrm{Im}(\mathbf{s}),\ \boldsymbol{\xi}_{\mathrm{s}} = -\frac{\mathrm{Re}(\mathbf{s})}{\boldsymbol{\omega}_{\mathrm{s}}} \tag{20}$$

2.3.2. Response calculations using Newmark's method

After incorporating the nonlinear vortex-excited force, the dynamic equation exhibits non-linearity in both velocity and displacement terms. To determine the structural displacement, a nonlinear Newmark's method should be employed [93]. It is based on a linearization of the equation for numerical integration in each time step, necessitating an iterative procedure to minimize the residual between the linearized results and the original nonlinear results. The partial derivatives of the residual contain tangent stiffness and damping matrices, which are derived from the nonlinear vortex-excited force model, as shown in Eqs. (21) and (22).

$$\begin{cases} \boldsymbol{g}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}}) = \mathbf{C}\cdot\dot{\boldsymbol{\eta}} + \mathbf{K}\cdot\boldsymbol{\eta} - \mathbf{F}_{\mathrm{VEF}}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}}) = \mathbf{C}\cdot\dot{\boldsymbol{\eta}} + \mathbf{K}\cdot\boldsymbol{\eta} - \mathbf{S}\cdot F_{\mathrm{VEF}}(\eta_i,\dot{\eta}_i) \\ \boldsymbol{K_T} = \partial g/\partial\boldsymbol{\eta} = \mathbf{K} - \partial\mathbf{F}_{\mathrm{VEF}}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}})/\partial\boldsymbol{\eta} = \mathbf{K} - \mathbf{S}\cdot\partial F_{\mathrm{VEF}}(\eta_i,\dot{\eta}_i)/\partial\eta_i \\ \boldsymbol{C_T} = \partial g/\partial\dot{\boldsymbol{\eta}} = \mathbf{C} - \partial\mathbf{F}_{\mathrm{VEF}}(\boldsymbol{\eta},\dot{\boldsymbol{\eta}})/\partial\dot{\boldsymbol{\eta}} = \mathbf{C} - \mathbf{S}\cdot\partial F_{\mathrm{VEF}}(\eta_i,\dot{\eta}_i)/\partial\dot{\eta}_i \end{cases} \tag{21}$$

The partial derivative in the above formula is calculated by the following equation:

$$\begin{cases} \partial F_{\mathrm{VEF}}(\eta_i,\dot{\eta}_i)/\partial\eta_i = \rho U^2 D\begin{bmatrix} a_2\int_0^l \phi_i^3(x)dx\frac{1}{D}\frac{\eta_i}{|\eta_i|} + 2a_3\int_0^l \phi_i^4(x)dx\left(\frac{1}{D}\right)^2\eta_i \\ +3a_4\int_0^l \phi_i^5(x)dx\left(\frac{1}{D}\right)^3\frac{\eta_i^3}{|\eta_i|} + 4a_5\int_0^l \phi_i^6(x)dx\left(\frac{1}{D}\right)^4\eta_i^3 \end{bmatrix}\frac{\dot{\eta}_i}{U} \\ \partial F_{\mathrm{VEF}}(\eta_i,\dot{\eta}_i)/\partial\dot{\eta}_i = \rho U^2 D\begin{bmatrix} a_1\int_0^l \phi_i^2(x)dx + a_2\int_0^l \phi_i^3(x)dx\left|\frac{\eta_i}{D}\right| + a_3\int_0^l \phi_i^4(x)dx\left(\frac{\eta_i}{D}\right)^2 \\ +a_4\int_0^l \phi_i^5(x)dx\left|\frac{\eta_i}{D}\right|^3 + a_5\int_0^l \phi_i^6(x)dx\left(\frac{\eta_i}{D}\right)^4 \end{bmatrix}\frac{1}{U} \end{cases} \tag{22}$$

where $\boldsymbol{K}_T$ and $\boldsymbol{C}_T$ represent the tangent stiffness and damping matrix.

2.4. Global optimization for the multiple tuned mass dampers

Global optimization is used to find the optimal parameters of the MTMDs, including the mass, damping ratio, stiffness and positions of the TMDs. The objective function is defined as the VIV amplitude of all modes with weights, as shown in Eq. (23). This function is based on an emphasis on the design specifications of completely suppressing VIV or reducing the VIV amplitude to a certain level.

$$J = \sum_i w_i A_i \tag{23}$$

$$A_i = \sqrt{2} \cdot \sqrt{\frac{\Sigma_N (\eta_{ss})^2}{N}} \cdot \max \phi_i \tag{24}$$

where A_i represents the maximum amplitude of the ith mode, w_i is a weight factor, η_{ss} represents the time series of modal displacement in the steady state, and N is the total count of data points in the time series.

The weight factors can be used if reducing the response of some modes is more critical than others. For instance, if passenger comfort is a top priority, modes with higher frequencies are assigned higher weight factors. On the other hand, if the risk of VIV is the primary concern, modes with lower frequencies are given higher weight factors since VIV occurs more often at low frequencies. In this study, the response of all modes is considered equally important, and all weights are therefore set to one. In this study, a bridge is employed as a case study. Fig. 3 shows a flowchart of the global optimization approach utilized throughout this work. The starting point is to obtain the structural and aerodynamic properties of the bridge. How many modes are susceptible to vortex-induced vibration depends on the basic reference wind speed at the bridge site. The number of modes to consider in the response calculation is also a critical parameter, which will be discussed in Section 3.4. It is widely known that the performance of a TMD improves when its mass increases. Therefore, the total mass of the TMDs and the number of TMDs are vital parameters that mainly depend on the project budget and specific circumstances and should be predetermined. Performing global optimization of multiple TMDs is challenging since it involves many parameters. Each iteration can be time-consuming, particularly when nonlinear time series calculations are necessary for each iteration. Many parameter combinations may give an equally good result, and finding the global minimum of the objective function might be challenging. This work aims to achieve a better solution than a mode-by-mode design approach and focus less on finding the global minimum of the objective function. In the present study, Particle Swarm Optimization (PSO) is employed as the computational framework for identifying the optimal parameters of MTMDs. Given the unique characteristics of the objective function for this specific problem, it is observed that certain parameter combinations may result in pronounced peaks, while other combinations exhibit a more plateau-like behavior. The PSO algorithm is particularly well-suited for this optimization task, owing to its inherent capabilities for parallel computing, which potentially enhance its efficacy in navigating the solution space. It should be noted, however, that PSO is not the exclusive optimization algorithm applicable to this problem. Consequently, it is advisable to explore a diverse range of optimization algorithms to ascertain the most effective approach for similar but distinct optimization challenges. The optimization process can be executed using MATLAB's Global Optimization Toolbox.

3. Multi-tuned mass damper control: Effects of quantity and placement

3.1. A multi-span continuous bridge

This study employs a six-span non-navigational continuous bridge from the Shen-Zhong Link as a case study, as depicted in Fig. 4. Two parallel girders rest on the same pier without any regular connections. The bridge consists of six spans, each 110 meters long, resulting in a total length of 660 m. Each girder is a typical hexagonal box design, with a height of 4 meters and a width of 20 m. The spacing between the two bridges varies from 0.5 meters to 6.7 m. This study considers a spacing of 6.7 meters since significant VIV was observed in the wind tunnel tests for configuration. Due to the fierce winds induced by typhoons and the narrow channel effect of the Peral River Estuary, the basic wind speed at the bridge site can reach 45.09 m/s [94–98]. The basic wind speed at the bridge site is defined as the annual maximum wind speed, averaged over a 10-minute interval, measured at a height of 10 meters above the ground (or water surface), with a 100-year return period [99]. In this study, geometric nonlinearity is not incorporated into the calculations because the amplitude of vortex-induced vibration (VIV) is considerably small relative to the geometric scale of the bridge. This assumption is corroborated to be reasonable and acceptable in other analogous studies [100–103]. However, it is imperative to note that when the amplitude of the vibration is substantial, the interrelation of nonlinear aeroelastic vibration, aerostatic deformation, and geometric nonlinearity should be accounted for [104]. Moreover, this study does not address the long-term evaluation and accidental alterations of the bridge's structural parameters due to structural damage and aging, which may potentially contribute to TMD mistuning over time. Fortunately, observational studies have indicated that the frequencies of a long-span bridge remain relatively stable, with only a 0.21% deterioration observed over a decade [53]. This degree of mistuning can be accommodated by the robustness of the TMDs. Should there be significant changes in the structural parameters, it would be necessary to retune the TMDs to maintain optimal performance. Both of these scenarios are advised to be avoided and are beyond the scope of this study.

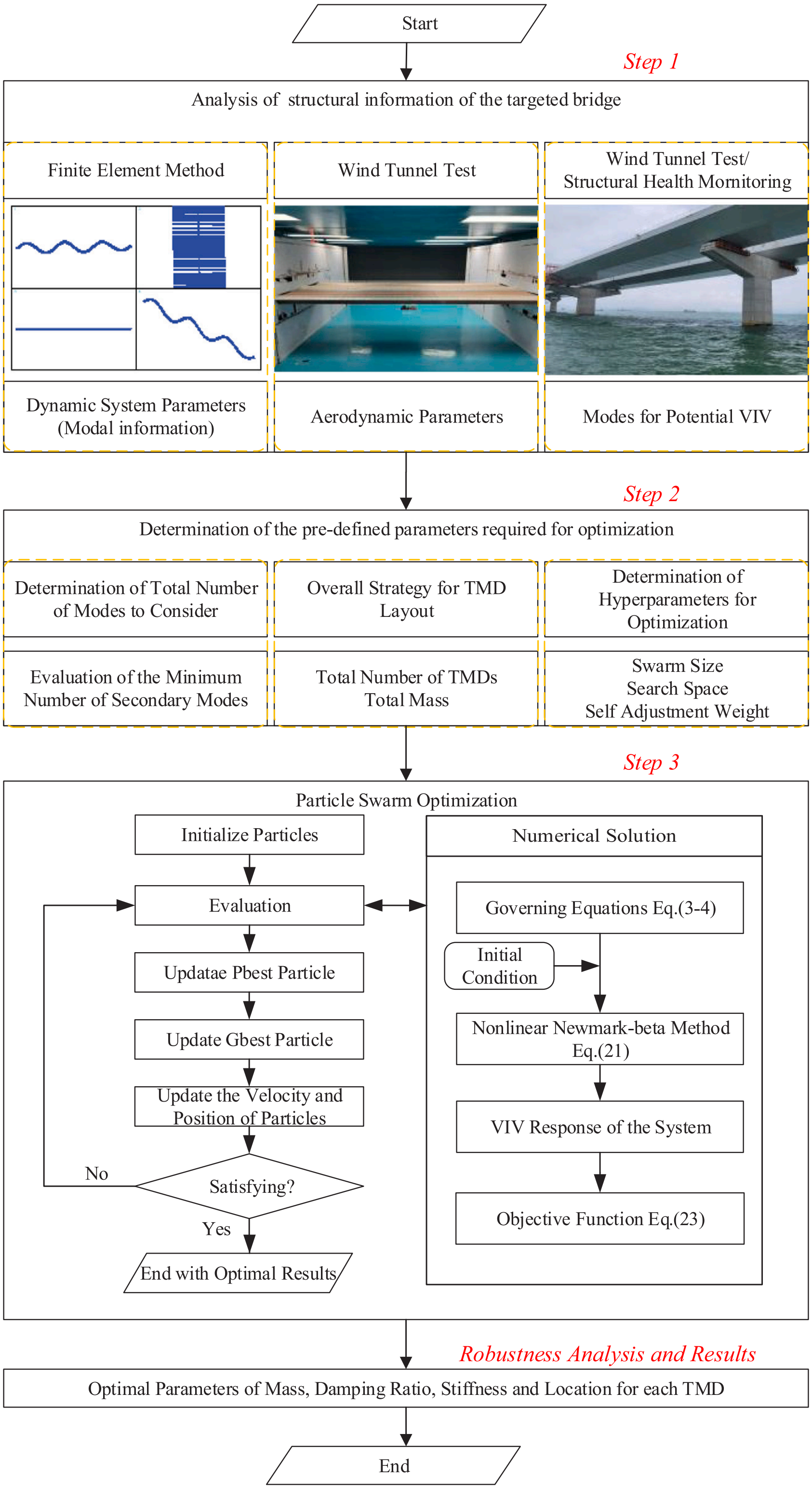


Fig. 3. The TMD optimization framework.

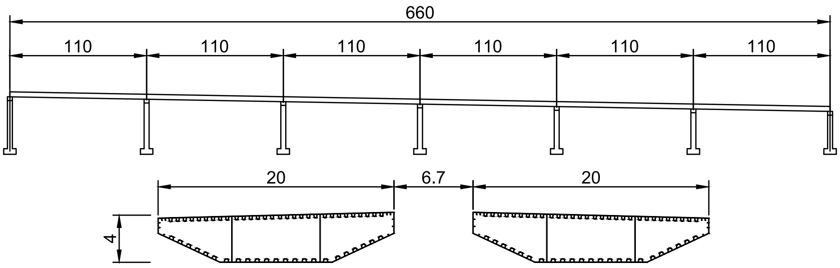
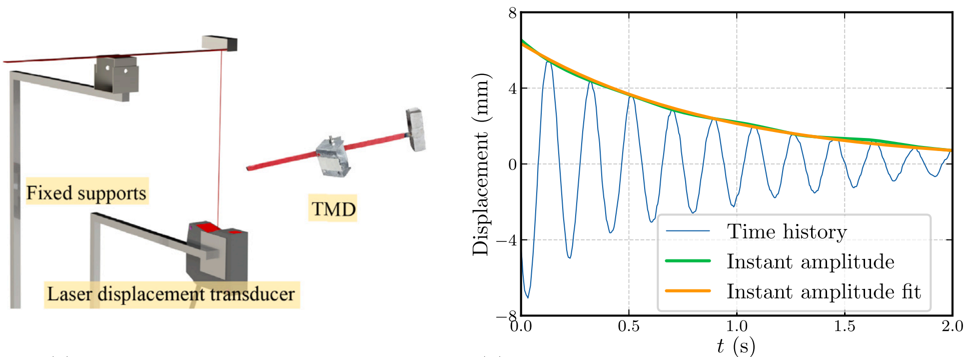


Fig. 4. Layout and cross-section of the non-navigational continuous bridge (unit: m).

Table 2
Design parameters of the SSSM system.

Parameter	Value
Vertical bending frequency (Hz)	5.25
Damping ratio (%)	0.3
Mass (kg/m)	22.2
Geometric scale ratio	1:30
Wind speed ratio	1:4.7
Slot width ratio (d/B)	0.335



(a) TMD and test equipment

(b) Time history of the free vibration of the TMD

Fig. 5. Parameter identification of TMD.

3.2. Wind tunnel test

Two rigid section models with a scale ratio of 1:30 are designed and manufactured to investigate the VIV characteristics, determine the aerodynamic parameters of the force model, and examine the TMD performance in suppressing VIV. Each section model is 3.6 meters long, 0.67 meters wide, and 0.13 meters high. The model's mass (m) is 22.2 kg/m, and the first vertical mode's natural frequency (f_h) is 5.25 Hz. Table 2 summarizes the parameters of the spring-suspended section model (SSSM).

Each TMD comprises a 250-gram iron block, an adjustable-length steel sheet, and tapes around the steel sheet serving as the mass, stiffness, and damping elements for the TMD. Due to the limitations in adjusting the damping ratio of TMDs, achieving precise adjustments is not feasible, and the TMD frequencies may experience variations resulting from adjustment errors. It should be noted that in real-world scenarios, the situation is distinct. The parameters of the TMDs can be tuned more easily to a desired value due to varying realizations of stiffness and dampers. Conversely, obtaining accurate structural information of the bridge proves to be a challenging task [53], a factor that may potentially lead to the mistuning of TMD. As a result of these limitations, the TMDs are not intended to determine the optimal parameters for designing the TMDs for actual bridges. Instead, the goal is to test the TMDs' performance by measuring the change in VIV amplitude after TMD installation and comparing it with the performance of the numerical calculation to validate the calculation method and the applicability of the force model. Four TMDs are installed on the suspended beams to mitigate the VIV for each sectional model. The parameters of the TMDs are calculated based on the free vibration time history captured by a laser displacement transducer, as shown in Fig. 5. The damping ratio of the TMDs is assumed to be linear and can be obtained by fitting the logarithmic decrement of the instant amplitude. The frequencies of the TMDs are calculated using the Fourier transform. Table 3 lists the parameters of the eight TMDs installed at the windward and leeward girders.

The experiments were conducted at Tongji University's TJ-3 boundary layer wind tunnel (see Fig. 6), which has dimensions of 14 meters in length, 15 meters in width, and 2 meters in height. The wind velocity in the wind tunnel is continuously adjustable

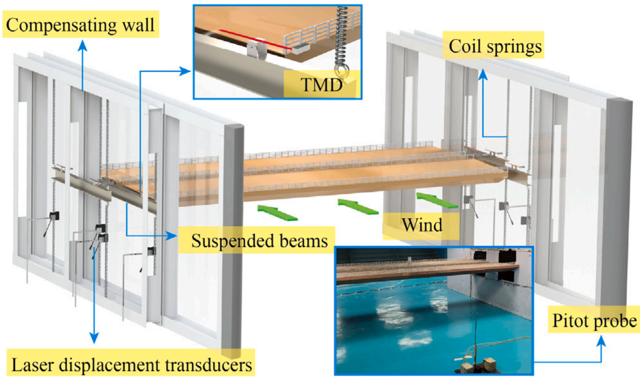


Fig. 6. SSSM, TMD and measurement system in wind tunnel.

Table 3
Parameters of TMD.

Girder of installation	Mass (kg)	Damping ratio (%)	Frequency (Hz)
Windward	0.25	4.2	5.16
		3.5	5.64
		3.4	5.46
		6	5.56
Leeward	0.25	8.2	5.22
		3.3	5.13
		6.1	5.23
		5.4	5.25

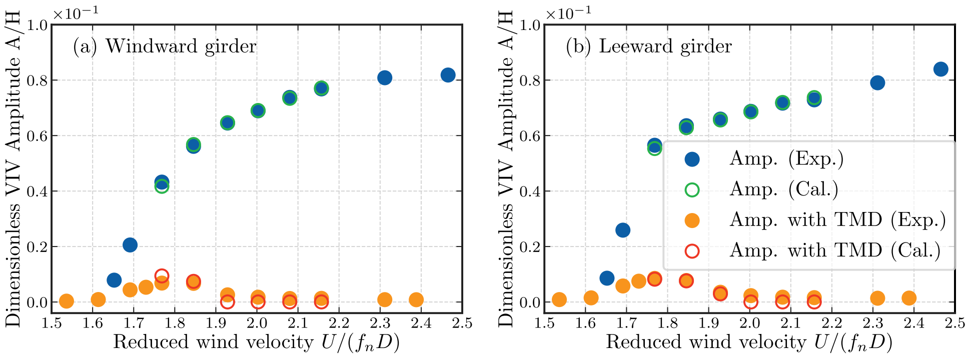


Fig. 7. Comparison of numerical calculation and experimental results. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

between 1 m/s and 17.6 m/s. The turbulence intensity and average airflow deviation angle in the wind tunnel are less than 2% and 0.2°, respectively. Additionally, the maximum blockage ratio at the angle of attack (AOA) of +3° is 8.4%, which is within the reasonable range for VIV, specifically within 10%, as Choi and Kwon [105] suggested.

The section models are suspended in two separate spring-suspended systems, which enable them to vibrate independently. The gap between the section models and compensating walls is less than 10 mm to avoid three-dimensional airflow at the end of the model. Each model is suspended using eight coil springs over and under the model (4 in parallel for each side), connected to two horizontal beams outside the compensating wall. The displacement signals of the models are recorded using eight Panasonic HL-C235CE-W laser displacement transducers (Osaka, Japan, the measuring range of ±200 mm) installed under the suspension arms. The signals are then processed by a data acquisition system (DASP3060S) with a maximum time delay of 15 ms per channel. Wind speed is measured synchronously using a pitot probe installed in front of the sectional model between the compensating walls. The displacement signal is sampled at 256 Hz, which is higher than the predominant vibration frequency during VIV experiments.

The experimental results indicate that the two girders exhibit a significant VIV amplitude at an angle of attack of +3°, as shown by the solid blue scatter in Fig. 7. After the installation of TMDs, only slight vibrations are observed at the beginning of the VIV, and the vibration in the rest of the lock-in region is completely suppressed.

Due to the nonlinear aerodynamic damping, the parameters of the polynomial model are determined by fitting the damping ratio curve with the amplitude, as calculated using Eq. (25). The amplitude-dependent damping ratio is calculated using the instant

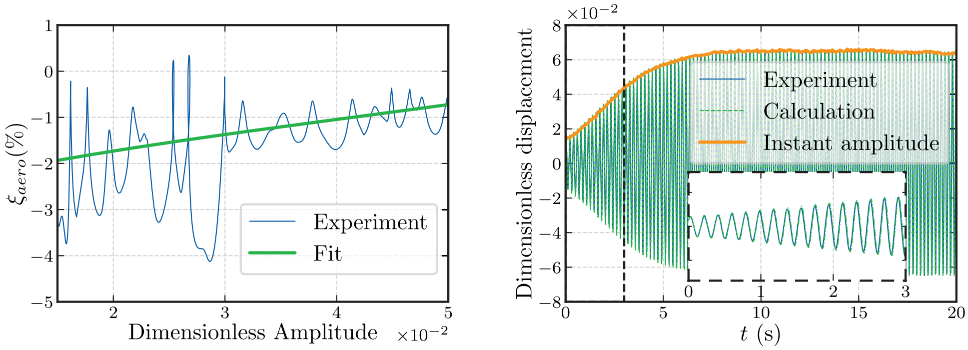


Fig. 8. Aerodynamic damping ratio at $U_r = 1.93$.

amplitude of the diverge-to-coverage time history and the process for identifying the parameters is detailed in the paper by Zhang et al. [92]. Fig. 8 depicts the identification process and the corresponding reconstructed time histories. In the enlarged subfigure, it is evident that the time history calculated using the identified parameters corresponds well with the experimental results. This indicates that the polynomial model can accurately predict the aerodynamic damping ratio at various amplitudes.

$$\xi_{aero}(A, K) = \frac{\rho U D}{2\omega_0 m}\left(a_1 + a_2\frac{4A}{3\pi} + a_3\frac{A^2}{4} + a_4\frac{8A^3}{15\pi} + a_5\frac{A^4}{8}\right) \tag{25}$$

3.3. VIV of a single mode with two tuned mass dampers

Controlling multi-mode VIV in a structure exhibiting multiple modes requires multiple TMDs, thereby engendering complexity. To facilitate a comprehensive comprehension of this matter, the following analysis is simplified into three steps. First, in Section 3.3, an investigation is taken to discern the influence of two TMDs with different parameters on VIV performance, considering a single mode. Next, Section 3.4 explores the evolution of control performance when one TMD is installed into a two-mode system. Lastly, an examination is conducted in Section 3.5 to determine the minimal number of modes required using the contribution value of the secondary modes to ascertain precise VIV calculation with installing TMD.

3.3.1. Effects of additional TMD on VIV performance

The scenario of one mode with two TMDs having different parameters is a practical situation where one TMD is designed for one mode, and the second TMD is designed for the other modes. However, the latter inevitably affects the first mode. To simulate this situation, the first mode of the six-span non-navigational continuous bridge is considered and the optimal parameters for the TMD are determined using the subsequent equations [99]. The first TMD has a mass ratio of 2%, a damping ratio of 0.086, and a natural frequency of 0.818 Hz. It is installed at the location with the largest modal shape of mode 1, where $\phi_{\mathrm{TMD1}} = 1$.

$$\mu_m = \frac{m_0\phi_i^2(x_0)}{\int_0^L m(x)\phi_i^2(x)dx} = 2\% \tag{26a}$$

$$\xi_{TMD1} = \sqrt{\frac{3\mu_m}{8(1+\mu_m)}} = 0.086 \tag{26b}$$

$$f_{TMD1} = \frac{1}{1+\mu_m}\cdot f_s = 0.818 \tag{26c}$$

TMD 2 has the same mass and damping ratio as TMD 1. The frequency ratio of TMD 2 to TMD 1 ranges from 0.7 to 1.4, and TMD 2 is installed at various locations on the bridge, where $\phi_{\mathrm{TMD2}} \in [0, 1]$. Therefore, the ratio of the modal shapes at the locations of TMD 2 and TMD 1 ranges from 0 to 1.

Fig. 9 displays the contour plots of the primary structure and the TMD amplitude after installing TMD 2. The dashed red line represents the amplitude with only TMD 1 installed. If the parameters of TMD 2 lie on this line, TMD 2 has no effect on VIV amplitude. In region 1 of Fig. 9a, TMD 2 has a positive effect on VIV control as it reduces vibration amplitude. This is particularly evident when the parameters of TMD 2 are close to TMD 1 in the area around point 1 in Fig. 9a, TMD 2 serves as an additional mass to TMD 1. The most interesting aspect is that TMD 2 can also reduce TMD 1's performance when its parameters are in region 2 or region 3. Comparing these two regions, it can be noted that when the frequency of TMD 2 is lower than TMD 1, TMD 2 is more likely to reduce TMD 1's performance, as region 3 is much larger than region 2. Fortunately, the contour lines are relatively sparse, indicating that the region where TMD 2 may exert a negative impact is quite limited. Therefore, although TMD 2 may decrease TMD 1's performance, it will not significantly increase VIV amplitude. It is worth noting that the vibration amplitude of

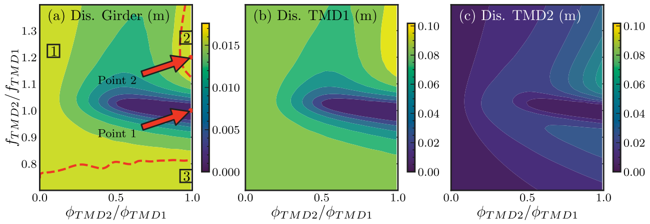


Fig. 9. Contour plots of primary structure and TMD amplitude after 2nd TMD installation.

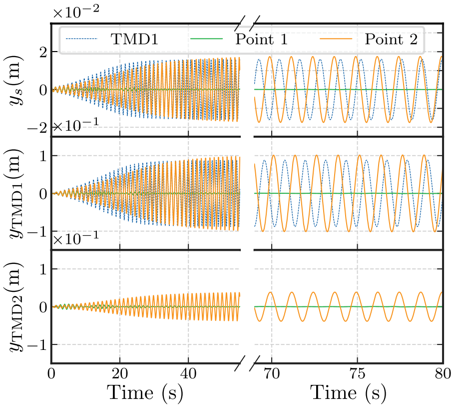


Fig. 10. Displacement histories of the bridge-damper system.

TMD 2 is much lower than that of TMD 1. However, the presence of TMD 2 will slightly change the modal characteristics of the force-structure-MTMDs system, which may detune TMD 1.

The case where only TMD 1 is installed is the reference testing case. Subsequent considerations include two configurations, wherein TMD 2 which is characterized by parameters corresponding to points 1 and 2 depicted in Fig. 9a is installed. Fig. 10 shows the displacements of the girder (y_s), TMD 1 (y_{TMD1}) and TMD 2 (y_{TMD2}) for the three different damper configurations.

It is demonstrated that installing only TMD 1 may not fully suppress VIV, and that installing TMD 2 with parameters corresponding to pint 1 can eliminate the vibration. When the properties of TMD 2 correspond to point 2, the final VIV amplitude is slightly larger than that with only TMD 1. However, during the onset of VIV, it develops much slower than that with only TMD 1. This suggests that the second TMD, even if not optimally tuned, can increase the damping ratio of the system when the amplitude is small. Conversely, when the amplitude is large, it may reduce the damping ratio compared to the one-TMD configuration. Different amplitudes in VIV correspond to different aerodynamic damping ratios. Generally, the negative aerodynamic damping ratio is much smaller for small amplitudes than for large amplitudes. As a result, the second TMD can increase the damping ratio of the system when the system is unstable. Still, it may affect TMD 1's performance and decrease the damping ratio of the system when the aerodynamic damping ratio is closer to zero, thus providing a larger steady-state response.

3.3.2. Effects of additional TMD on aerodynamic damping ratio

The proposed hypothesis concerning the impact of TMD 2 on the aerodynamic damping ratio is sought to be validated. This involves an exploration of how the aerodynamic damping ratio alters with varying amplitudes and parameters of TMD 2. Since the vortex-excited force can be converted into amplitude-dependent aerodynamic damping, manually setting a system damping ratio ξ_s (the sum of aerodynamic and mechanical damping ratios) enables the structure to vibrate at a targeted amplitude. Consequently, the performance of the TMD at a specific amplitude can be evaluated. Fig. 11 presents contour plots illustrating the damping ratio originating from a single TMD at various system damping ratios, while Fig. 12 depicts a two-TMD case, where an optimized TMD 1 is pre-installed. The axes have been rendered dimensionless using the TMD 1 parameters. The red dashed line represents the contour line corresponding to a damping ratio increase of 0.01 and the yellow dotted-dashed line demarcates the regions of damping ratio increase or decrease. In the single-TMD case, installing a TMD does not appear to reduce the system damping ratio, irrespective of its parameters. Conversely, it indicates that higher system damping ratios result in diminished damping contributions from the TMD. For the two-TMDs case, installing TMD 2 in the bridge-TMD 1 system leads to a decrease in the system damping ratio. The region of decreased damping ratio expands with increasing system damping ratios. Additionally, no region in Fig. 12(b) and (c) exhibits a damping ratio increase of 1%. It is important to note that installing additional TMDs that are not optimally tuned or designed for other vibration modes may adversely influence pre-existing tuned TMDs, particularly in cases of large vibration amplitudes (i.e. high system damping ratio). As a result, ignoring the interaction between the TMDs in the design process may lead to inaccurate results and underestimate VIV amplitudes at the steady state.

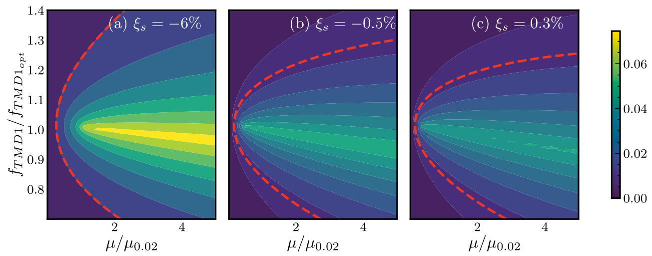


Fig. 11. Contour plots of TMD-induced damping ratio increase at various system damping ratios for one TMD.

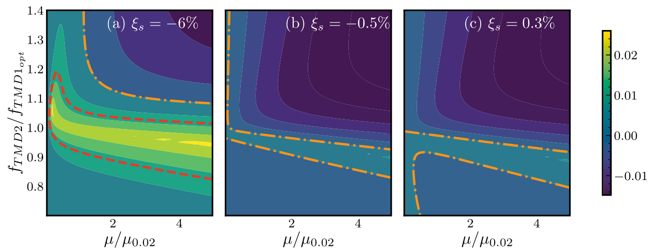


Fig. 12. Contour plots of TMD-induced damping ratio increase at various system damping ratios for two TMDs.

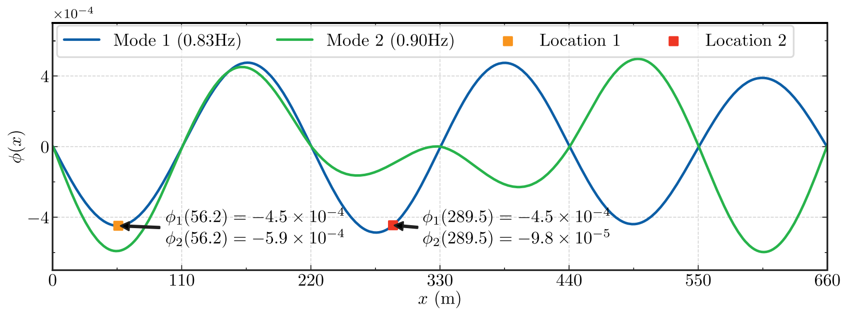


Fig. 13. Layout of tuned mass dampers with the same modal displacement in mode 1.

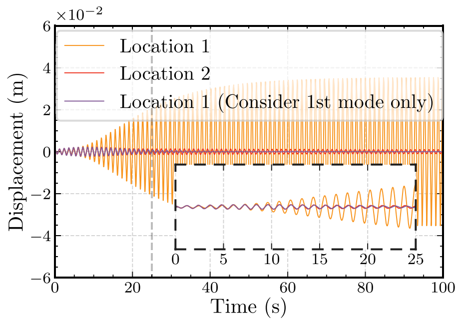


Fig. 14. Time histories of the displacement of the girder.

3.4. VIV in mode one of a two-mode system with one TMD

In addition to the interaction between multiple TMDs in one mode, the interaction between multiple modes needs to be considered when optimizing one TMD. Initially, a case study is undertaken to evaluate the influence of a TMD's location on the VIV amplitude. In this scenario, the bridge is modeled as a two-mode structure, excited in the first mode. A single TMD, tuned to the first mode of the bridge, is positioned at two distinct locations respectively. Both locations exhibit the same modal displacement in mode 1. The second modal displacement of location 1 is larger than that in location 2, as shown in Fig. 13.

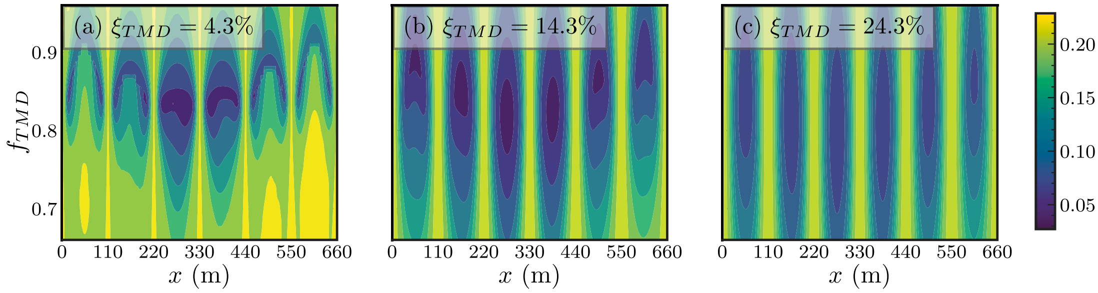


Fig. 15. Bridge deck displacement at various TMD locations and parameters considering two modes. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

The time histories of the girder's displacement, with the TMD at two distinct locations, are illustrated in Fig. 14. For comparison, the displacement of the girder, considering only a single mode with the TMD at location 1, is also depicted. Both cases demonstrate a reduction in VIV amplitude, yet the TMD positioned at location 2 outperforms the one at location 1. Furthermore, it is noteworthy that if only mode 1 is considered, the calculated displacement when TMD 1 is at location 1 is significantly smaller than in the two-mode system. This suggests that disregarding mode 2 could potentially lead to an underestimation of the VIV amplitude. This observation can be attributed to the pronounced mode coupling effect that occurs when a TMD is positioned at a location where the modal displacements for both mode 1 and mode 2 are large. The vibration of mode 2 is inevitably excited, acting as an additional spring attached to the TMD, which subsequently reduces its performance.

However, it may be necessary to install distributed TMDs on the bridge to avoid a single TMD with a large mass. Therefore, it may be inevitable to install some TMDs in places where the secondary mode is large. A parameter study is conducted to investigate further the consequence of placing the TMD in such locations. The VIV amplitudes for a system with two modes and one TMD with different frequencies and damping ratios at various locations are shown in Fig. 15. Three different damping ratios of the TMD are considered: 4.3%, 14.3% and 24.3%. The x-axis represents the location of the TMD, the y-axis is the frequency and the colormap represents the response of the system. The darkest area represents the optimal frequency and placement of the TMD, indicating the minimal vibration amplitude of the girder. It is evident that when the TMD is installed at the middle spans of the bridge where the modal displacement of mode 2 is minimal, the optimal frequency is close to the natural frequency of mode 1 as calculated by Eq. (26c). However, if the TMD is installed in the side spans, the optimal frequency is higher for all three damping ratios due to the influence of mode 2. Observations also indicate that the frequency needs to be tuned to a significantly higher value when positioned in the middle of the side spans, compared to when it is near the extremities of these spans. Similar to the mode-by-mode analysis result, a larger damping ratio of the TMD shows a more robust performance in the sense that it works well for a wider frequency range.

3.5. Contribution value of the secondary modes

The presence of secondary modes may affect the optimal parameters of the TMD, and ignoring them may lead to inaccurate prediction of the response of the bridge and the damper. The accuracy of the response prediction improves as more modes are considered, but adding modes also increases the complexity. Therefore, a balance should be found between the number of secondary modes considered and the complexity of the problem. Fig. 16 shows the displacement of the girder when considering one to five modes. The parameters of the TMD are obtained by Eq. (26). Results considering the first five modes of the bridge are plotted for reference. It is evident that when the TMD is installed near the piers it does not contribute much to damp the vibrations and the effect of including more modes is insignificant. However, when the TMD is installed in the middle of one span, the influence of the secondary modes is pronounced. Compared to considering only mode 1, further consideration of mode 2 will make the displacement calculation of the side and second spans more accurate. When mode 3 is also considered, it increases the accuracy of the displacement in the middle spans. Including mode 4 and mode 5 does not improve the accuracy significantly, but some small changes are observed. Therefore considering only the first vibration modes may be sufficient to assess the performance of a TMD to suppress VIV. A further investigation is conducted below into the number of vibration modes that should be considered.

Figs. 17–22 show the contribution of secondary modes to the girder displacement response. The wind-resistant design code for China and wind tunnel test results indicate that the first six modes of the bridge may be set in VIV. This is because the higher modes exceed the bridge reference wind speed at 56.47 m/s [99]. Vortex-excited forces are thus applied to each of the first six modes to evaluate the TMD's performance and to investigate the influence of secondary modes. Six TMDs are designed using Eq. (26) and placed where the modal displacement is the largest. Fig. 17 shows the calculated displacements of the girder including 1–8 modes when the vortex-excited force is applied to mode 1. The black bold line in the figure represents the displacement of the girder, considering 100 modes and can be regarded as the converged result. The other lines represent the displacements considering an increasing number of modes. The displacement converges quickly, and only a few modes are needed to achieve accurate results. The other subfigures in the figure show the percentage contribution of the modes not directly set in VIV. In each subfigure, the horizontal represents the mode shape displacement at the TMD for the mode set in VIV, and the y-axis represents the mode shape

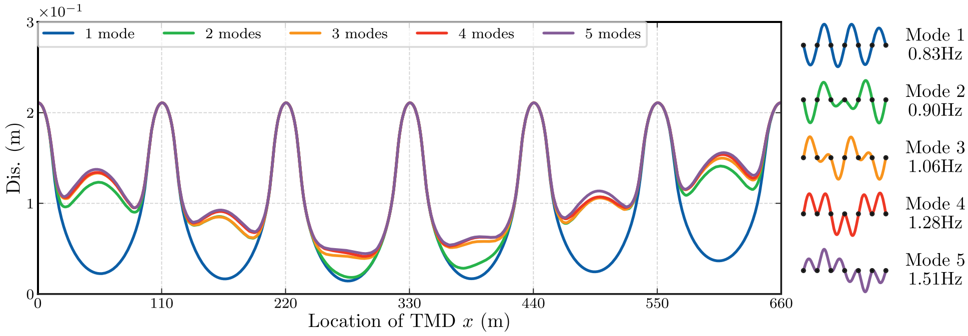


Fig. 16. Layout of tuned mass dampers with the same modal displacement in mode 1.

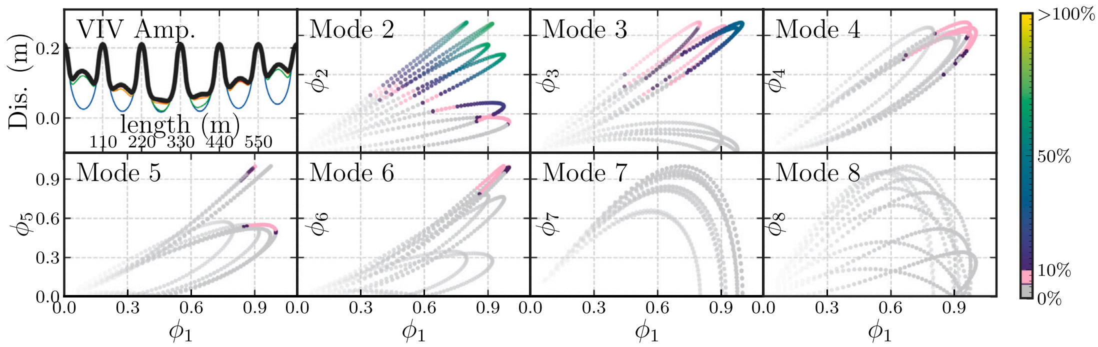


Fig. 17. Contribution values of the secondary modes (VEF on mode 1). (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

displacement at the TMD for the mode not set in VIV, but that might contribute to the response. Points, where the secondary mode contributes more than 10% to the total response, are highlighted using a gradient color, while points, where the secondary mode contributes between 5% to 10% to the total response, are shown in pink. Points with a gray color indicate that the secondary mode contributes less than 5% to the total response. The transparency of each point in the subfigures illustrates the system's total response when considering 100 modes. Points with high transparency mean that the VIV response is low, while points that are not transparent mean that the VIV response is large. Fig. 17 shows that mode 2 significantly contributes to the total response. When the TMD is installed where the modal displacements of modes 1 and 2 are both greater than 0.3, there are significant multimode effects, and it is not sufficient to consider only one mode in the calculation. Similarly, in Fig. 17, mode 3 shows an important contribution when the TMD is installed in locations where the modal displacements of modes 1 and 3 are both greater than 0.6. Secondary modes 4, 5, and 6 exhibit a median contribution when the TMD is installed in locations where the modal displacements of the VIV mode and the secondary modes are all large. On the other hand, regardless of where the TMD is installed, modes 7 and 8 have no significant impact on the total response. The result when vortex-excited force is applied to mode 2 is shown in Fig. 18. As with the result when the force is applied to mode 1, if the TMD is installed in locations where the modal displacement of the main mode and the secondary modes are both large, the secondary modes contribute significatly. Figs. 19 to 22 shows that despite the difference in frequency of modes 3 to 6 being more significant than modes 1 and 2, the neighbor modes contribute significantly to the total response. Additionally, it is worth noting that the location where the secondary modes do not significantly influence the TMDs performance is always where the VIV amplitude is low. To summarize, one needs to consider the first 10 modes of the bridge when calculating the VIV response in the first six modes to ensure accurate response predictions.

4. Optimization of multiple tuned mass dampers

4.1. Control scheme and comparison

Four control schemes are proposed to compare the mode-by-mode design and explore the potential for reducing the number of TMDs. In this study, the total mass ratio of the TMDs is set to 2%, which is within the suggested limit of 4% [106]. The calculated total mass of the TMDs is $\sum m_j = 200.154$ tons, which is within the same order of magnitude as TMDs used in practical applications [76]. Scheme 1 involves the mode-by-mode design, where six TMDs are used to control each mode of VIV. The TMD for a given mode is installed in a location where the modal displacement of the corresponding mode is largest, and the parameters of these TMDs are calculated by Eq. (26). Scheme 2 involves optimizing the mass, damping ratio, stiffness, and location of the TMDs

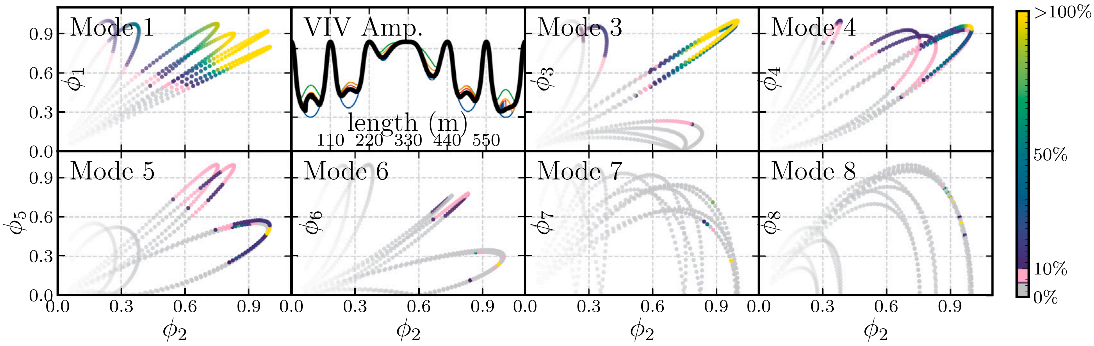


Fig. 18. Contribution values of the secondary modes (VEF on mode 2).

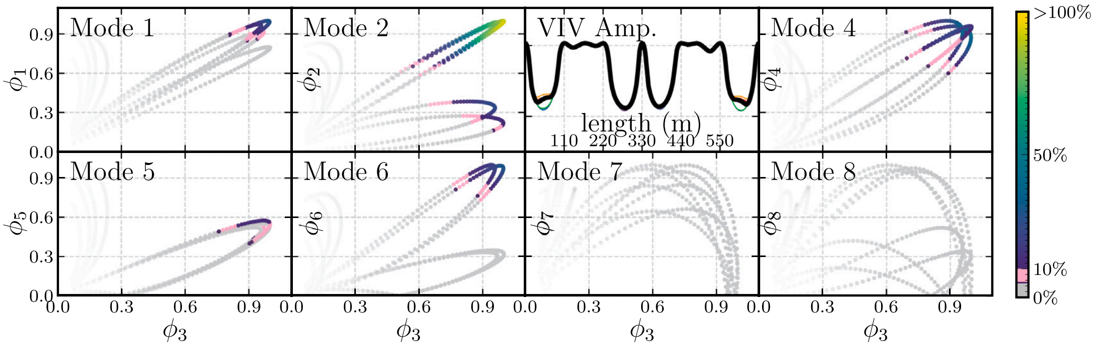


Fig. 19. Contribution values of the secondary modes (VEF on mode 3).

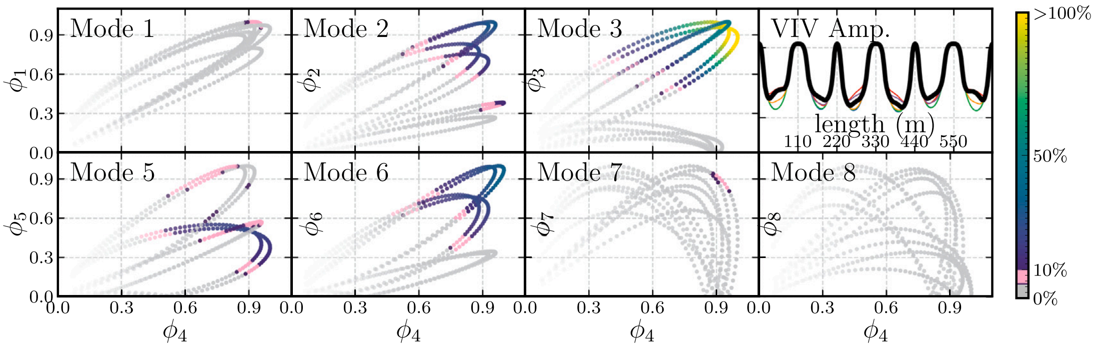


Fig. 20. Contribution values of the secondary modes (VEF on mode 4).

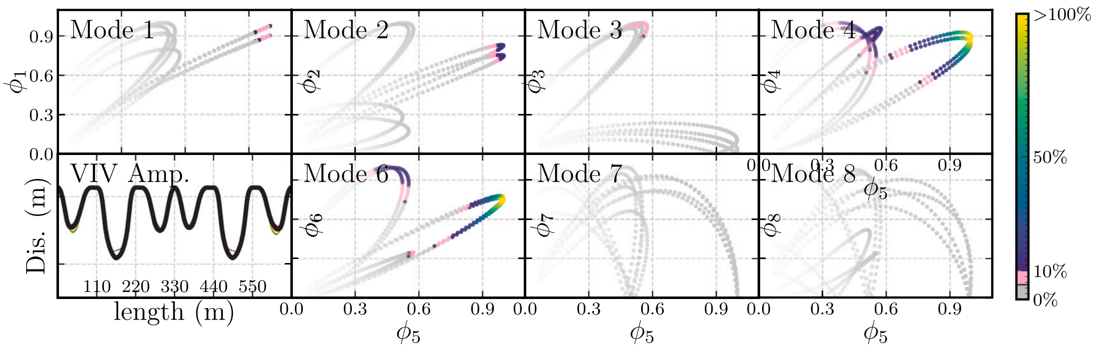


Fig. 21. Contribution values of the secondary modes (VEF on mode 5).

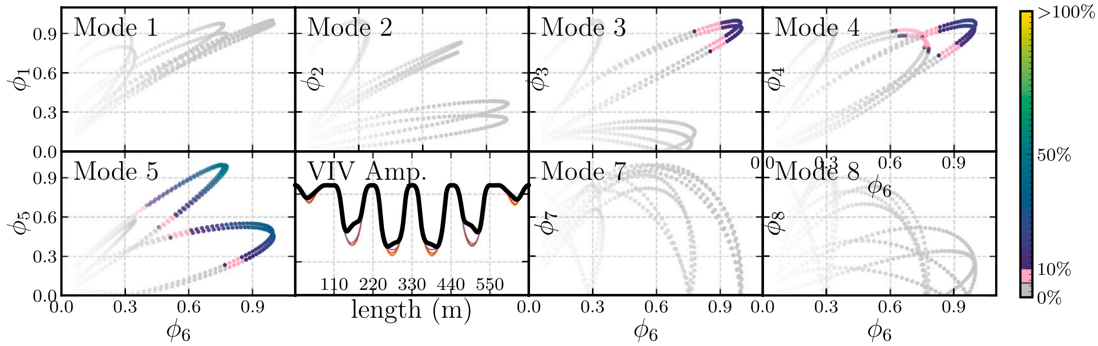


Fig. 22. Contribution values of the secondary modes (VEF on mode 6).

Table 4
Constraints and lateral limits of design variables.

	Scheme 1	Scheme 2	Scheme 3	Scheme 4
Number of TMDs	$n = 6$	$n = 6$	$n = 5$	$n = 3$
Mass (kg)	$m_j = \frac{1}{n} m_b \times 2\%$	$\sum_{j=1}^{n} m_j = m_b \times 2\% = n \cdot \overline{m}$ $m_{j=1\cdots n-1} \in [0.5\overline{m}, \overline{m}]$		
Frequency (Hz)	Based on Eq. (26)	$f_j \in [0.7, 2]$		
Damping ratio (%)		$\xi_j \in [5, 20]$		
Location (m)	$x_j = \mathrm{argmax}_l \phi_j(l)$	$x_j \in [0, 660]$		

where m_b is the total mass of the bridge, and $\overline{m}$ is the average mass of the TMDs.

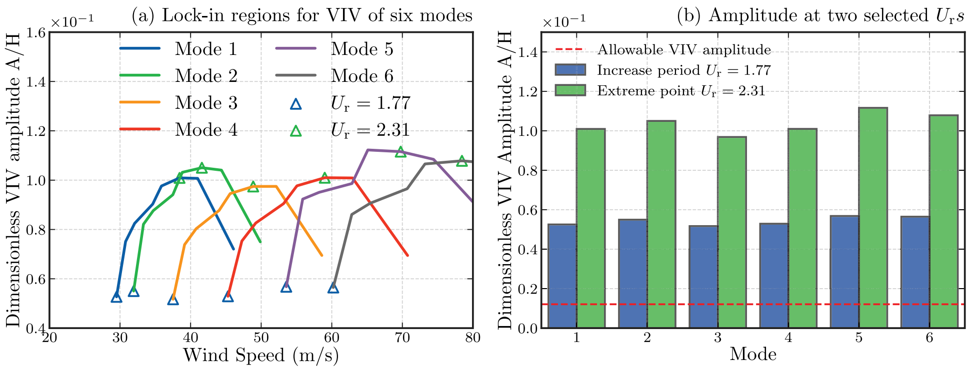


Fig. 23. Amplitude of VIV at the first six modes.

simultaneously by particle swarm optimization. Scheme 3 uses the same optimization algorithm as scheme 2 but reduces the number of TMDs to five to control six VIV-prone modes. Finally, scheme 4 proposes a further reduction of the number of TMDs to three. A comprehensive list of the constraints and lateral limits of the design variables for each scheme is provided in Table 4.

The lock-in regions of the first six modes of VIV and the amplitude of each mode are shown in Fig. 23. It is reasonable that the layout of TMDs should be effective for VIV at different reduced wind speeds. Therefore, the subsequent optimization is based on the increase period to determine the optimal TMD parameters, with the results double-checked by the extreme point to ensure robustness to different reduced wind speeds. It is noted that the negative aerodynamic damping ratio of the increase period is even smaller than that of the extreme point at small amplitudes. Thus, the increase period with the reduced wind speed of $U_r = 1.77$ is the worst-case condition for TMD design, as demonstrated in Fig. 7.

4.2. Optimized parameters of multiple tuned mass dampers

The optimized parameters of the TMDs targeting the four schemes are plotted in Figs. 24 and 25, and detailed in Table 5. For scheme 1 and scheme 2, each TMD is placed at the generalized coordinate of the corresponding mode, as shown in Fig. 24. The size of the square represents the mass of the TMD. Comparing the mode-by-mode design and the optimized parameters of the TMDs, it is noted that the frequencies and locations for TMDs targeting modes 2, 3, 5, and 6 change only slightly. The mass of the four TMDs decreases by about one-fourth from 33 tons to approximately 25 tons. The reduced mass is added to the TMD targeting mode 4, which increases from 33 tons to 67 tons due to the modal displacement of the installation location of TMD 4 being smaller than

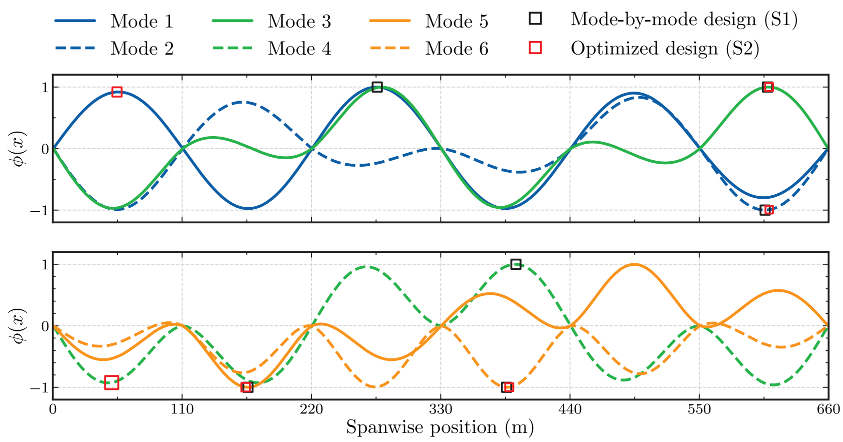


Fig. 24. Layout of the TMDs in Scheme 1 and 2.

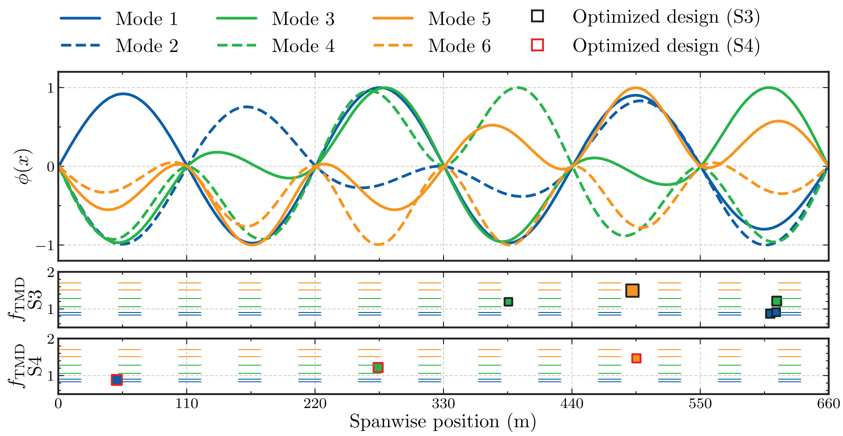


Fig. 25. Layout of the TMDs in Scheme 3 and 4. (For interpretation of the references to color in this figure legend, the reader is referred to the web version of this article.)

its original location. This also causes the frequency of the TMD targeting mode 4 to change more than that of the other TMDs. Specifically, the frequency of the TMD targeting mode 4 is reduced by 0.1 Hz compared to the traditional design, changing from 1.27 Hz to 1.17 Hz. TMD 1 is transferred from the middle span to the side span with minor changes in the other parameters. In reality, it is challenging to identify a location where only the modal displacement for a certain mode is the largest. A change in the parameters of one TMD may affect the performance of all the other TMDs, making manual tuning challenging. In scheme 2, the location change of TMDs 1 and 4 and the mass redistribution of all TMDs are automatically determined by the optimization algorithm and almost reduce the objective function by a factor of two demonstrating better performance.

Fig. 25 displays the layouts of the other schemes where the number of TMDs is reduced. The color of the square represents the closest modal frequencies compared to the girder. The y-axis represents the frequencies of the TMDs, while the horizontal axis represents their locations. In scheme 3, the four TMDs with green and blue colors exhibit the characteristics of a single TMD for its targeted mode, placed at a large modal displacement of the corresponding mode. Even though each of the first four TMDs is designed to control one mode, the frequencies of the TMDs are not exactly the same as in the traditional design. Interestingly, the frequency of TMD 3 is slightly higher than that of mode 3, while the frequency of TMD 4 is slightly lower than that of mode 4. This results in the frequencies of the two TMDs being closer to each other, but they are not located at the same position. The remaining TMD has a double mass, intended to control the VIV of modes 5 and 6. However, its frequency is closer to mode 5 than mode 6, suggesting that it may perform better in controlling the VIV of mode 5. In scheme 4, neither of the TMD frequencies is close to any modal frequency of the girder. The frequencies of the first two lower frequency TMDs are between modes 1 and 2, and modes 3 and 4, respectively, aiming to control two modes with a single TMD. The last TMD is assigned a frequency close to mode 5, similar to scheme 3. The findings indicate that the optimized TMDs with reduced numbers do not enhance the control performance of the sixth mode when the objective is to regulate the overall performance of the first six modes. However, if there are particular requirements for controlling the sixth mode, for instance, if the sixth mode is deemed most critical, the weight coefficient for the sixth mode can be adjusted to a higher value in the objective function. Moreover, multi-objective optimization can be employed to tailor the TMDs to meet specific requirements.

Table 5
Optimized design parameters for each control scheme.

Control scheme		Parameters				Displacement (m)	U_r=1.77	U_r=2.31
		Mass (kg)	Location (m)	Frequency (Hz)	Dampingratio (%)			
S1	TMD1	33359	276	0.83	5.0	Mode 1	0.056	0.046
	TMD2	33359	606	0.90	4.0	Mode 2	0.046	0.003
	TMD3	33359	608	1.06	4.0	Mode 3	0.022	0.000
	TMD4	33359	394	1.27	5.0	Mode 4	0.053	0.026
	TMD5	33359	166	1.50	5.0	Mode 5	0.040	0.002
	TMD6	33359	386	1.69	5.0	Mode 6	0.007	0.000
	–	–	–	–	–	**Sum**	0.224	0.077
S2	TMD1	33324	55	0.85	5.0	Mode 1	0.037	0.002
	TMD2	22068	609	0.90	14.6	Mode 2	0.002	0.000
	TMD3	24441	609	1.02	5.0	Mode 3	0.001	0.000
	TMD4	67079	50	1.17	13.2	Mode 4	0.035	0.002
	TMD5	28502	164	1.51	5.0	Mode 5	0.019	0.000
	TMD6	24742	388	1.69	5.0	Mode 6	0.022	0.000
	–	–	–	–	–	**Sum**	0.116	0.004
S3	TMD1	32965	610	0.87	8.4	Mode 1	0.056	0.046
	TMD2	29714	615	0.91	14.1	Mode 2	0.002	0.000
	TMD3	27376	386	1.19	17.5	Mode 3	0.012	0.000
	TMD4	37915	615	1.21	9.4	Mode 4	0.039	0.008
	TMD5	72186	492	1.49	8.6	Mode 5	0.001	0.000
	–	–	–	–	–	Mode 6	0.052	0.034
	–	–	–	–	–	**Sum**	0.163	0.088
S4	TMD1	79590	50	0.88	9.8	Mode 1	0.013	0.000
	TMD2	66492	274	1.22	7.1	Mode 2	0.004	0.000
	TMD3	54074	495	1.47	8.9	Mode 3	0.025	0.000
	–	–	–	–	–	Mode 4	0.071	0.067
	–	–	–	–	–	Mode 5	0.001	0.000
	–	–	–	–	–	Mode 6	0.051	0.033
	–	–	–	–	–	**Sum**	0.166	0.100

Based on the results of the three optimized schemes, it is observed that redistributing mass and relocating some TMDs are the most significant changes when the number of TMDs remains consistent with that in the mode-by-mode design. However, when the number of TMDs is less than the modes prone to VIV, our findings suggest that TMDs with relatively low mass are suitable for controlling the VIV of one mode, while those with a relatively high mass are ideal for controlling the VIV of multiple modes.

4.3. Control efficiencies

Table 5 presents the value of the objective function. Control efficiency is defined as the reduction in amplitude after installing TMDs, calculated as $(1 - J_{\mathrm{TMD}}/J_{\mathrm{non-TMD}}) \times 100\%$. A higher control efficiency indicates a more optimal TMD design. Figs. 26 and 27 shows the control efficiency at two reduced wind speeds. For $U_r = 1.77$, all three optimized TMD designs outperform the mode-by-mode design in the overall control efficiency for the sum of six modes. Scheme 2 performs better in the first five modes, but there is a slight decrease in control efficiency for the sixth mode due to the reduced mass of TMD6 compared to the mode-by-mode design. Scheme 3 also demonstrates better control performance in the first five modes, but the sixth mode's control effect is not as good as that of scheme 2 because there is no TMD dedicated to the sixth mode in scheme 3, yet it still suppresses more than 75% of the VIV amplitude. Scheme 4 exhibits a noticeable decrease in control efficiency for mode 4 and mode 6 compared to scheme 2, possibly because the reduction in the number of TMDs does not account for all modes. Nevertheless, scheme 4 still yields better overall performance than the mode-by-mode design. If TMDs are unable to fully control all modes of VIV, the control effect of specific modes may be diminished, and the weight of the objective function can be adjusted to prioritize the control of certain modes.

As discussed in Section 4.1, the negative aerodynamic damping at $U_r = 2.31$ is less severe than at $U_r = 1.77$. Consequently, Fig. 27 shows that all schemes exhibit better control efficiency. Scheme 2 still yields the highest overall control efficiency, although the difference is less significant than at $U_r = 1.77$ since nearly all schemes have a control efficiency of over 95%.

4.4. Robustness to structure frequency and damping ratio

Besides control efficiency, the robustness of the TMD network design is also an important criterion. During a bridge's lifespan, environmental factors may cause changes in frequency and damping ratio, necessitating the evaluation of the schemes' effectiveness in controlling VIV under different structural conditions [21,22,69]. The variation in overall control performance and control performance for each mode is illustrated in Figs. 28 and 29, respectively.

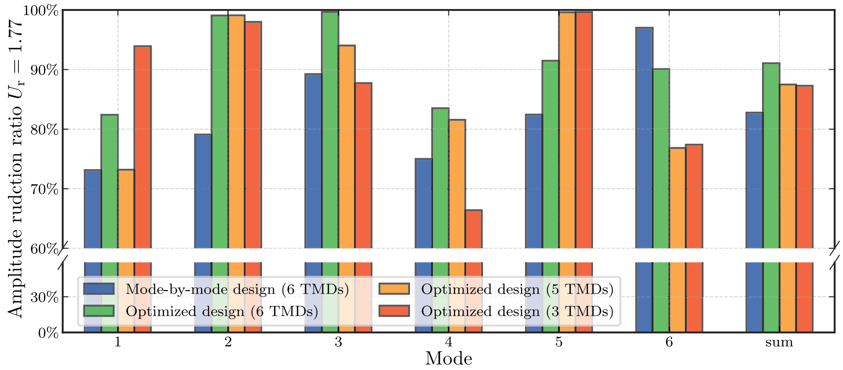


Fig. 26. Amplitude reduction by using different schemes (U_r = 1.77).

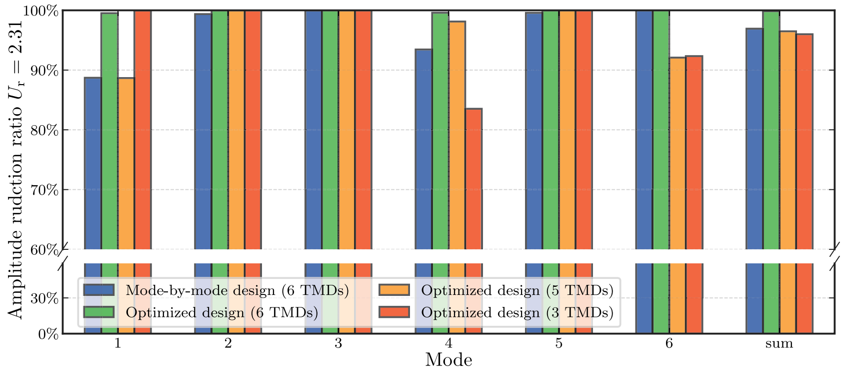


Fig. 27. Amplitude reduction by using different schemes (U_r = 2.31).

Fig. 28 displays the amplitude variation with frequency for the four schemes. All schemes exhibit the lowest amplitudes when there is no variation in structural frequency for overall performance. Scheme 1, scheme 2, and scheme 4 exhibit similar overall performance trends as the frequency changes. Scheme 3 demonstrates better performance when the structural frequency decreases but worse performance when it increases. No scheme has a significant disadvantage in terms of robustness to frequency variation. For control performance in each mode, the minimum amplitude might not always be realized at a 0% variation of structural frequency, particularly for scheme 1. This is because although the TMD frequencies in scheme 1 are well-tuned for each mode, not considering the secondary modes' effects may cause the TMDs to be mistuned. A similar phenomenon also occurs in schemes 2, 3, and 4 when controlling the fourth mode because neighboring TMDs help suppress the fourth mode when the frequency of the structure changes. The decreasing of the structural frequency for mode 6 can also increase control performance for these schemes due to neighboring TMDs aiding in suppressing the sixth mode, whereas increasing the structural frequency results in all TMDs becoming mistuned for the sixth mode. Fig. 29 illustrates the amplitude variation with damping ratio for the four schemes. All schemes exhibit a similar trend in overall performance and the performance for each mode, with amplitude decreasing as the damping ratio increases. This is due to the fact that the negative amplitude-dependent aerodynamic damping ratio is a monotonically increasing function of the amplitude. As the damping ratio of the structure increases, the negative damping ratio must decrease to maintain a zero damping ratio at the VIV onset, resulting in a decrease in amplitude. Consequently, there is no significant difference in robustness to structural frequency and damping ratio among the four schemes. The performance of the optimized TMD design remains robust to changes in structural frequency and damping ratio. However, despite the robustness of the TMD design to variations in structural frequency and damping ratio, it is imperative to meticulously determine the structural frequencies and damping ratios to guarantee that the TMDs are optimally tuned [107]. Furthermore, the application of robust optimization techniques, which take into account the uncertainties associated with structural parameters but were not implemented in this study, could serve as a valuable approach to enhance the robustness of the TMD design further [51,108].

5. Conclusions

The present paper proposes a framework for optimizing the parameters for a network of TMDs to control VIV of a structure with closely spaced modes considering a multimode approach. The governing equations for the force-structure-MTMDs system are derived and the nonlinear vortex-excited force is adopted. The nonlinear equations are solved by time integration, while a wind

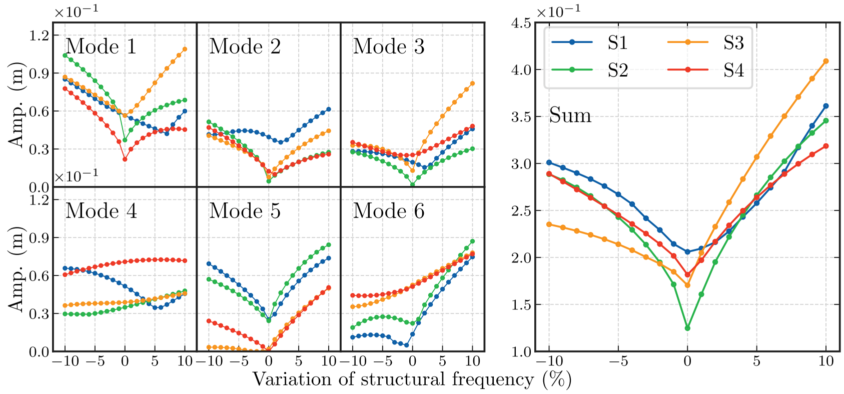


Fig. 28. Sensitivity of the TMD against structural frequency.

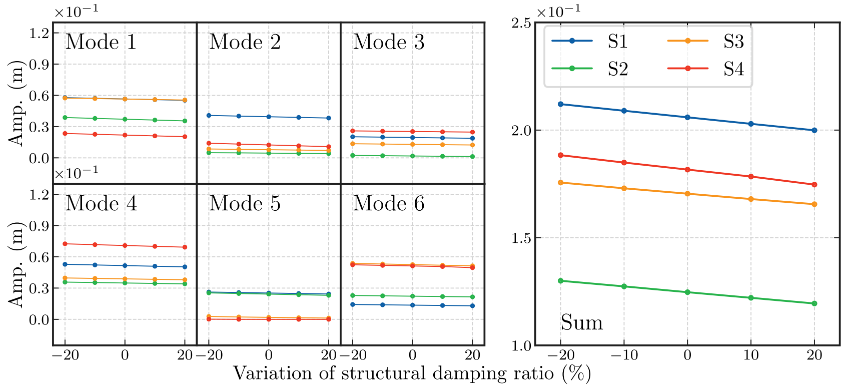


Fig. 29. Sensitivity of the TMD against structural damping ratio.

tunnel test is conducted to validate the results. A global optimization algorithm is used to optimize the parameters of the TMDs, and a case study of a six-span continuous bridge is used to demonstrate the effectiveness of the proposed framework. The robustness of the optimized TMD design is also examined. Several conclusions can be summarized:

1. The wind tunnel test shows that the polynomial nonlinear vortex-excited force model can effectively model the vibration characteristics of the system including TMDs.
2. Installing more TMDs on a structure having a TMDs can make the existing TMDs untuned and reduce their performance significantly, particularly when the aerodynamic damping ratio is small.
3. Ignoring secondary modes can lead to errors in the calculation of TMD performance and VIV amplitude. A novel method is proposed to calculate the mode contribution and determine the minimum number of secondary modes to be involved.
4. The optimized parameters derived from the optimization algorithm demonstrate enhanced control performance compared to the traditional design, even when the same total mass of the TMDs is employed. Specifically, the optimized design achieves a 91.1% reduction in amplitude for the most adverse conditions, representing a 10% improvement over the traditional design. Furthermore, at the peak wind velocity corresponding to the maximum VIV amplitude, the optimized design achieves a 99.9% reduction in amplitude, indicating nearly a complete suppression of vibration.
5. If the number of TMDs installed is equal to the number of VIV-prone modes, the optimized parameters are similar to the ones obtained by a mode-by-mode approach. However, when the number of TMDs installed is less than the VIV-prone modes, one or more TMDs should control multiple modes, and their frequencies may not be close to either controlled mode.
6. The optimized design demonstrates no significant difference in robustness compared to the mode-by-mode design when the structure's parameters change, while still maintaining superior control performance.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

Acknowledgments

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